

Method of Images: Electric Field on a Grounded Conducting Plane (Consistency Check)

We now compute the electric field *just above* the conductor, at a point

$$P : (\rho, \phi, z = 0^+),$$

using Coulomb's law for the *image-charge configuration*, and show that it reduces to the same result already implicit in the Lecture-16 through $\mathbf{E} = -\nabla\phi$ and $\sigma = -\epsilon_0 \partial\phi/\partial z|_{z=0}$.

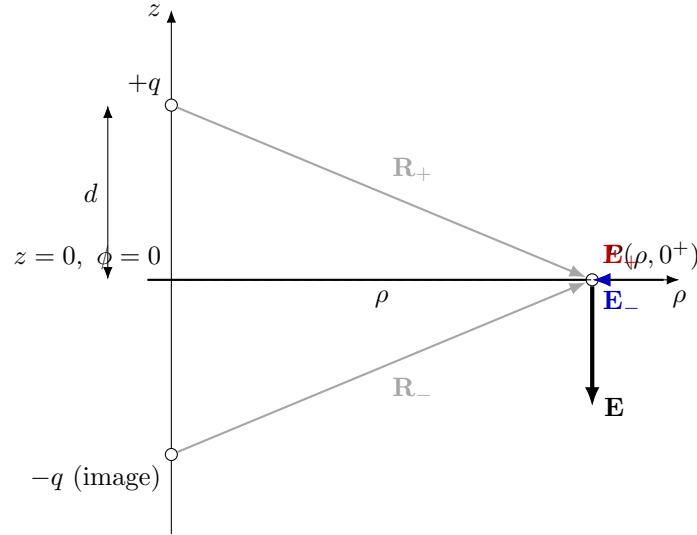


Figure 1: Field directions at $P(\rho, 0^+)$: $\mathbf{E}_+$ points away from $+q$ (along $+q \rightarrow P$), $\mathbf{E}_-$ points toward $-q$ (along $P \rightarrow -q$). The tangential components cancel, leaving a purely normal field.

(A) Coulomb-law derivation at $z = 0^+$

At $P(\rho, 0^+)$, the displacement vectors from the charges to P are

$$\mathbf{R}_+ = \rho \hat{\rho} - d \hat{z}, \quad \mathbf{R}_- = \rho \hat{\rho} + d \hat{z}, \quad |\mathbf{R}_\pm| = \sqrt{\rho^2 + d^2}.$$

Hence, using Coulomb's law $\mathbf{E} = \frac{1}{4\pi\epsilon_0} q \frac{\mathbf{R}}{R^3}$,

$$\mathbf{E}_+(P) = \frac{1}{4\pi\epsilon_0} q \frac{\rho \hat{\rho} - d \hat{z}}{(\rho^2 + d^2)^{3/2}},$$

$$\mathbf{E}_-(P) = \frac{1}{4\pi\epsilon_0} (-q) \frac{\rho \hat{\rho} + d \hat{z}}{(\rho^2 + d^2)^{3/2}}.$$

Adding:

$$\mathbf{E}(P) = \mathbf{E}_+(P) + \mathbf{E}_-(P) = \frac{q}{4\pi\epsilon_0} \frac{(\rho \hat{\rho} - d \hat{z}) - (\rho \hat{\rho} + d \hat{z})}{(\rho^2 + d^2)^{3/2}} = -\frac{q}{4\pi\epsilon_0} \frac{2d}{(\rho^2 + d^2)^{3/2}} \hat{z}.$$



So the field on the plane is purely normal and points *toward* the conductor (negative $\hat{z}$).

Physical meaning (why the $\hat{\rho}$ part cancels)

On a perfect conductor in electrostatics, the tangential electric field must be zero at the surface. In the image picture, the two contributions at $z = 0$ have equal and opposite $\hat{\rho}$ components, so only the normal component survives.

(B) Showing this is a special case of the general $\mathbf{E}(\rho, z)$

For any point (ρ, z) in the physical region $z > 0$, the image-charge field is

$$\mathbf{E}(\rho, z) = \frac{q}{4\pi\epsilon_0} \left[\frac{\rho \hat{\rho} + (z-d)\hat{z}}{(\rho^2 + (z-d)^2)^{3/2}} - \frac{\rho \hat{\rho} + (z+d)\hat{z}}{(\rho^2 + (z+d)^2)^{3/2}} \right], \quad z > 0.$$

Setting $z = 0^+$ immediately gives

$$\mathbf{E}(\rho, 0^+) = -\frac{q}{4\pi\epsilon_0} \frac{2d}{(\rho^2 + d^2)^{3/2}} \hat{z},$$

which is exactly the “plane” expression derived above.

Expression via $\mathbf{E} = -\nabla\phi$. From the Lecture-16 we already had (for $z > 0$)

$$\phi(\rho, z) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{\sqrt{\rho^2 + (z-d)^2}} - \frac{q}{\sqrt{\rho^2 + (z+d)^2}} \right),$$

so

$$E_z(\rho, z) = -\frac{\partial\phi}{\partial z}.$$

Evaluating at $z = 0$,

$$\left. \frac{\partial\phi}{\partial z} \right|_{z=0} = \frac{q}{4\pi\epsilon_0} \frac{2d}{(\rho^2 + d^2)^{3/2}} \implies E_z(\rho, 0^+) = -\frac{q}{4\pi\epsilon_0} \frac{2d}{(\rho^2 + d^2)^{3/2}},$$

which matches the Coulomb-law result above.

(C) Bonus consistency: $\sigma(\rho) = \epsilon_0 E_z(0^+)$

Since $E_{\text{inside}} = 0$ for a conductor, the normal boundary condition gives

$$\sigma(\rho) = \epsilon_0 E_z(\rho, 0^+) = -\epsilon_0 \left(\frac{q}{4\pi\epsilon_0} \frac{2d}{(\rho^2 + d^2)^{3/2}} \right) = -\frac{q d}{2\pi(\rho^2 + d^2)^{3/2}},$$

exactly the induced surface charge density obtained (Lecture-16).



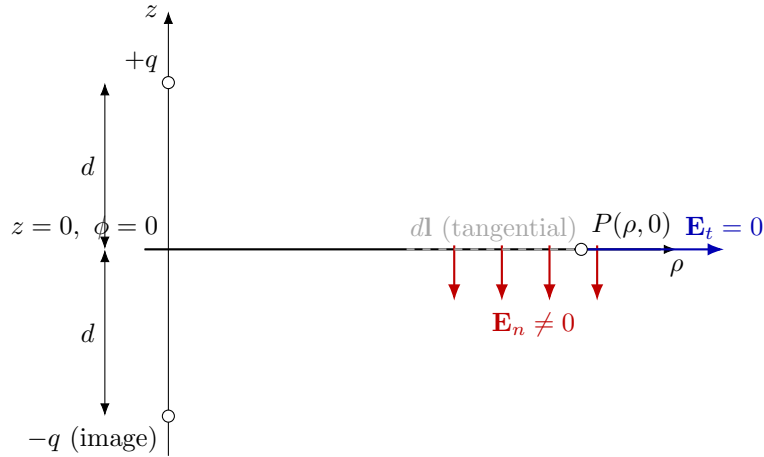


Figure 2: On a perfect conductor, the surface field has no tangential component: $\mathbf{E}_t = 0$. Hence along any path lying on the conductor surface, $\mathbf{E} \cdot d\mathbf{l} = 0$.

Why $\phi(\rho, 0) = 0$ on a grounded infinite plane

In electrostatics the potential difference between two points A and B is defined by

$$\phi(B) - \phi(A) = - \int_A^B \mathbf{E} \cdot d\mathbf{l}.$$

We choose the usual reference $\phi(\infty) = 0$ (valid here since the field of a localized charge distribution vanishes at infinity).

Potential is constant on the conductor. Take any two points P_1, P_2 on the plane $z = 0$. Choose a path entirely on the plane from P_1 to P_2 . Along this path $d\mathbf{l}$ is tangential to the surface, while $\mathbf{E}$ at the surface is purely normal ($\mathbf{E}_t = 0$). Therefore,

$$\phi(P_2) - \phi(P_1) = - \int_{P_1}^{P_2} \mathbf{E} \cdot d\mathbf{l} = 0,$$

so ϕ is *constant* over the entire conducting plane.

Grounding fixes that constant to zero. If the plane is grounded, its potential is defined to be zero. Hence the constant value must be 0, i.e.

$$\boxed{\phi(\rho, 0) = 0 \quad \text{for all } \rho}.$$

Uniqueness $\Rightarrow$ equivalence with the image-charge problem

In the physical region $z > 0$, the potential must satisfy Laplace/Poisson equation with the boundary condition $\phi(\rho, 0) = 0$ and $\phi \rightarrow 0$ as $r \rightarrow \infty$. The image-charge construction produces a potential that



satisfies these same conditions (by construction). By the **uniqueness theorem**, the solution in $z > 0$ is therefore the *same* for:

- a point charge $+q$ above a grounded infinite conducting plane, and
- two point charges $+q$ at $z = d$ and $-q$ at $z = -d$ in free space (restricted to $z > 0$).

Conclusion

Basically, a complex boundary-value problem is brought down to a simpler free-space charge configuration, which we can solve directly using Coulomb's law by invoking the uniqueness theorem and the method of images.

Two-dielectric parallel-plate capacitor: derivation of $\mathbf{D}$, $\mathbf{E}$, and $\phi(x)$

Consider two infinite parallel plates of area A , separated by distance d . The bottom plate at $x = 0$ is grounded ($\phi = 0$), and the top plate at $x = d$ is held at $\phi = V_0$. The region $0 < x < x_0$ is filled with dielectric ε_2 , and the region $x_0 < x < d$ is filled with dielectric ε_1 . Assume 1D field (fringing neglected), so $\mathbf{E} = E_x(x) \hat{\mathbf{x}}$ and $\mathbf{D} = D_x(x) \hat{\mathbf{x}}$.

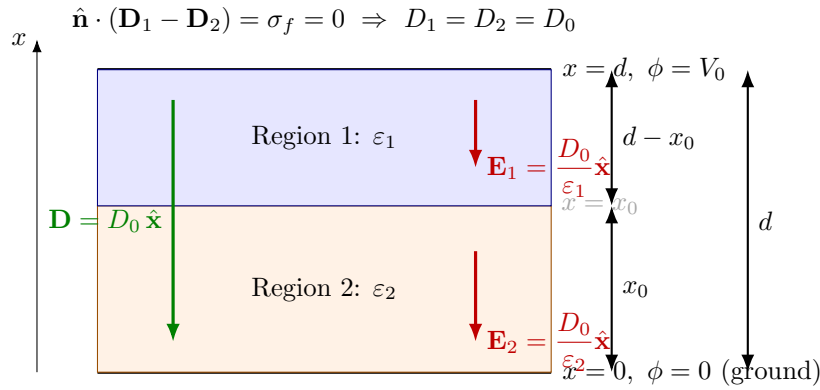


Figure 3: Parallel-plate capacitor partially filled with two dielectrics. There is no free charge at the interface, hence D_n is continuous and $\mathbf{D}$ is constant. The electric field is piecewise constant: $\mathbf{E} = \mathbf{D}/\varepsilon$.

Step 1: Use Gauss law for $\mathbf{D}$ in each dielectric (no free *volume* charge). Inside the dielectrics there is no free charge density:

$$\rho_f = 0 \implies \nabla \cdot \mathbf{D} = 0.$$

With 1D dependence, $\nabla \cdot \mathbf{D} = \frac{dD_x}{dx} = 0$, hence D_x is constant in each region:

$$D_x = \begin{cases} D_2, & 0 < x < x_0, \\ D_1, & x_0 < x < d. \end{cases}$$

Step 2: Continuity of D_n across the interface (no free *surface* charge at $x = x_0$). Apply the boundary condition on the interface:

$$\hat{\mathbf{n}} \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \sigma_f^{(\text{interface})}.$$

Since the interface is free of free charge, $\sigma_f^{(\text{interface})} = 0$, so

$$D_1 = D_2 \equiv D_0.$$

Therefore $\mathbf{D}$ is the *same constant* in both layers:

$$\mathbf{D}(x) = \begin{cases} D_0 \hat{\mathbf{x}}, & 0 < x < x_0, \\ D_0 \hat{\mathbf{x}}, & x_0 < x < d. \end{cases}$$

Step 3: Obtain $\mathbf{E}$ from the constitutive law $\mathbf{D} = \varepsilon \mathbf{E}$. In each dielectric,

$$\mathbf{E} = \frac{\mathbf{D}}{\varepsilon}.$$

Hence,

$$\mathbf{E}(x) = \begin{cases} \frac{D_0}{\varepsilon_2} \hat{\mathbf{x}}, & 0 < x < x_0, \\ \frac{D_0}{\varepsilon_1} \hat{\mathbf{x}}, & x_0 < x < d. \end{cases}$$

(If you want the physically correct direction: for top plate at higher potential, $\mathbf{E}$ points downward; then write $\mathbf{E} = -\frac{D_0}{\varepsilon_i} \hat{\mathbf{x}}$ and keep the same magnitudes.)

Step 4: Derive $\phi(x)$ using $\mathbf{E} = -\nabla\phi$. In 1D, $E_x = -\frac{d\phi}{dx}$, so $\frac{d\phi}{dx} = -E_x$. Using the piecewise-constant E_x :

Region 2: $0 < x < x_0$ (dielectric ε_2).

$$\frac{d\phi}{dx} = -\frac{D_0}{\varepsilon_2} \quad \Rightarrow \quad \phi(x) = \phi(0) - \frac{D_0}{\varepsilon_2} x.$$

With $\phi(0) = 0$,

$$\phi(x) = -\frac{D_0}{\varepsilon_2} x, \quad 0 < x < x_0.$$

Region 1: $x_0 < x < d$ (dielectric ε_1).

$$\frac{d\phi}{dx} = -\frac{D_0}{\varepsilon_1} \quad \Rightarrow \quad \phi(x) = \phi(x_0) - \frac{D_0}{\varepsilon_1} (x - x_0).$$



But from Region 2,

$$\phi(x_0) = -\frac{D_0}{\varepsilon_2}x_0.$$

So

$$\phi(x) = -\frac{D_0}{\varepsilon_2}x_0 - \frac{D_0}{\varepsilon_1}(x - x_0), \quad x_0 < x < d.$$

Step 5: Use the plate boundary condition $\phi(d) = V_0$ to determine D_0 . Evaluate $\phi(d)$ from the Region 1 expression:

$$V_0 = \phi(d) = -\frac{D_0}{\varepsilon_2}x_0 - \frac{D_0}{\varepsilon_1}(d - x_0).$$

Thus (magnitude form),

$$D_0 = \frac{V_0}{\frac{x_0}{\varepsilon_2} + \frac{d - x_0}{\varepsilon_1}} \quad (\text{sign depends on chosen } \hat{\mathbf{x}} \text{ direction}).$$

Final piecewise results

$$\mathbf{D}(x) = D_0 \hat{\mathbf{x}}, \quad D_0 = \frac{V_0}{\frac{x_0}{\varepsilon_2} + \frac{d - x_0}{\varepsilon_1}}$$

$$\mathbf{E}(x) = \begin{cases} \frac{D_0}{\varepsilon_2} \hat{\mathbf{x}}, & 0 < x < x_0, \\ \frac{D_0}{\varepsilon_1} \hat{\mathbf{x}}, & x_0 < x < d, \end{cases} \quad \phi(x) = \begin{cases} -\frac{D_0}{\varepsilon_2}x, & 0 < x < x_0, \\ -\frac{D_0}{\varepsilon_2}x_0 - \frac{D_0}{\varepsilon_1}(x - x_0), & x_0 < x < d. \end{cases}$$

Uniqueness Theorem Applied: Two-Dielectric Parallel-Plate Capacitor (Alternate Method)

1) Governing equation $\Rightarrow \phi$ is linear in each region

Electrostatics gives

$$\mathbf{E} = -\nabla\phi, \quad \nabla \cdot \mathbf{D} = \rho_f.$$

Inside the dielectrics there is no free volume charge, so $\rho_f = 0$. With 1D dependence (only in x),

$$\nabla \cdot \mathbf{D} = \frac{dD_x}{dx} = 0.$$

In each homogeneous isotropic layer, $\mathbf{D} = \varepsilon\mathbf{E} = -\varepsilon\frac{d\phi}{dx}\hat{\mathbf{x}}$, hence

$$\frac{d}{dx}\left(\varepsilon\frac{d\phi}{dx}\right) = 0 \quad \Rightarrow \quad \frac{d^2\phi}{dx^2} = 0 \quad (\text{since } \varepsilon = \text{const. in each region}).$$

Therefore ϕ is linear in each region.



2) Solve ϕ in each region (determine constants)

Let Region I be $x_0 < x < d$ with permittivity ε_1 , and Region II be $0 < x < x_0$ with permittivity ε_2 .

Region I (ε_1):

$$\phi_1(x) = C_1x + C_2, \quad \phi_1(d) = V_0 \Rightarrow C_2 = V_0 - C_1d.$$

So

$$\boxed{\phi_1(x) = C_1x + (V_0 - C_1d), \quad x_0 < x < d.}$$

Region II (ε_2):

$$\phi_2(x) = C_3x + C_4, \quad \phi_2(0) = 0 \Rightarrow C_4 = 0,$$

so

$$\boxed{\phi_2(x) = C_3x, \quad 0 < x < x_0.}$$

3) Apply interface conditions at $x = x_0$

(i) Continuity of potential:

$$\phi_1(x_0) = \phi_2(x_0) \Rightarrow C_1x_0 + (V_0 - C_1d) = C_3x_0. \quad (\text{BC1})$$

(ii) Continuity of D_n (no free surface charge at interface):

$$\hat{\mathbf{n}} \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \sigma_f^{(\text{int})} = 0 \Rightarrow D_{1x} = D_{2x}.$$

But

$$\mathbf{E}_1 = -\frac{d\phi_1}{dx}\hat{\mathbf{x}} = -C_1\hat{\mathbf{x}}, \quad \mathbf{E}_2 = -\frac{d\phi_2}{dx}\hat{\mathbf{x}} = -C_3\hat{\mathbf{x}},$$

so

$$\mathbf{D}_1 = \varepsilon_1\mathbf{E}_1 = -\varepsilon_1C_1\hat{\mathbf{x}}, \quad \mathbf{D}_2 = \varepsilon_2\mathbf{E}_2 = -\varepsilon_2C_3\hat{\mathbf{x}}.$$

Thus $D_{1x} = D_{2x}$ gives

$$\boxed{\varepsilon_1C_1 = \varepsilon_2C_3} \Rightarrow \boxed{C_3 = \frac{\varepsilon_1}{\varepsilon_2}C_1}. \quad (\text{BC2})$$

4) Solve for C_1 and C_3

Substitute $C_3 = (\varepsilon_1/\varepsilon_2)C_1$ into (BC1):

$$\begin{aligned} C_1x_0 + V_0 - C_1d &= \frac{\varepsilon_1}{\varepsilon_2}C_1x_0 \\ \Rightarrow V_0 &= C_1 \left[d + x_0 \left(\frac{\varepsilon_1}{\varepsilon_2} - 1 \right) \right] = C_1 \left[d - \left(1 - \frac{\varepsilon_1}{\varepsilon_2} \right) x_0 \right]. \end{aligned}$$

Hence

$$\boxed{C_1 = \frac{V_0}{d - (1 - \varepsilon_1/\varepsilon_2)x_0}} \quad \text{and} \quad \boxed{C_3 = \frac{\varepsilon_1}{\varepsilon_2}C_1}.$$



5) Final expressions for $\mathbf{E}$, $\mathbf{D}$, and ϕ

$$\mathbf{E}(x) = \begin{cases} -C_3 \hat{\mathbf{x}}, & 0 < x < x_0, \\ -C_1 \hat{\mathbf{x}}, & x_0 < x < d, \end{cases} \quad \mathbf{D}(x) = \begin{cases} -\varepsilon_2 C_3 \hat{\mathbf{x}}, & 0 < x < x_0, \\ -\varepsilon_1 C_1 \hat{\mathbf{x}}, & x_0 < x < d. \end{cases}$$

and

$$\phi(x) = \begin{cases} C_3 x, & 0 < x < x_0, \\ C_1 x + (V_0 - C_1 d), & x_0 < x < d. \end{cases}$$

Consistency with the previous-session form (using D_0)

Define the (constant) displacement magnitude

$$D_0 \triangleq \varepsilon_1 C_1 = \varepsilon_2 C_3 \quad (\text{so } \mathbf{D} = -D_0 \hat{\mathbf{x}}).$$

From $\phi(d) - \phi(0) = V_0 = -\int_0^d E_x dx$ with E_x piecewise constant,

$$V_0 = \left(\frac{D_0}{\varepsilon_2} \right) x_0 + \left(\frac{D_0}{\varepsilon_1} \right) (d - x_0) \Rightarrow D_0 = \frac{V_0}{\frac{x_0}{\varepsilon_2} + \frac{d - x_0}{\varepsilon_1}}.$$

Then $C_3 = D_0/\varepsilon_2$ and $C_1 = D_0/\varepsilon_1$, giving exactly the earlier piecewise results:

$$\phi(x) = \begin{cases} \frac{D_0}{\varepsilon_2} x, & 0 < x < x_0, \\ \frac{D_0}{\varepsilon_2} x_0 + \frac{D_0}{\varepsilon_1} (x - x_0), & x_0 < x < d, \end{cases} \quad \mathbf{E} = \begin{cases} -\frac{D_0}{\varepsilon_2} \hat{\mathbf{x}}, & 0 < x < x_0, \\ -\frac{D_0}{\varepsilon_1} \hat{\mathbf{x}}, & x_0 < x < d. \end{cases}$$

Iterative use of the uniqueness theorem (forward–backward idea)

1. **Guess the unknown interface potential:** assume $\phi(x_0) = \phi_0$ (unknown constant).
2. **Forward step (from $x = 0$ to $x = x_0$):** Solve Laplace's equation in Region II with Dirichlet data $\phi(0) = 0$ and $\phi(x_0) = \phi_0$. By the **uniqueness theorem**, this solution is unique, hence E_2 and D_2 in Region II are uniquely fixed by ϕ_0 :

$$\phi_2(x) = \frac{\phi_0}{x_0} x, \quad E_2 = -\frac{d\phi_2}{dx} = -\frac{\phi_0}{x_0}, \quad D_2 = \varepsilon_2 E_2.$$

3. **Interface transfer (at $x = x_0$):** No free charge at the interface implies $D_1 = D_2$. Therefore the field in Region I is uniquely determined:

$$D_1 = D_2 \quad \Rightarrow \quad E_1 = \frac{D_1}{\varepsilon_1}.$$

4. **Forward continuation (from $x = x_0$ to $x = d$):** Using the same guessed ϕ_0 , solve Laplace's equation in Region I with $\phi(x_0) = \phi_0$. This gives a unique $\phi_1(x)$, hence it predicts a *computed* top-plate value $\phi_1(d)$.
5. **Backward check (from $x = d$ back to $x = x_0$):** But the top plate boundary condition is fixed: $\phi(d) = V_0$. So we compare the predicted value $\phi_1(d)$ with the required value V_0 . If $\phi_1(d) \neq V_0$, our assumed ϕ_0 is wrong.
6. **Update and iterate:** Adjust ϕ_0 and repeat the forward–backward sweep:

$$x = 0 \rightarrow x_0 \rightarrow d \quad (\text{predict } \phi(d)) \quad \text{and} \quad d \rightarrow x_0 \rightarrow 0 \quad (\text{enforce } \phi(d) = V_0).$$

This process converges to the *one* value of ϕ_0 that makes the boundary condition at $x = d$ exactly satisfied.

7. **Why the final ϕ_0 is unique:** If two different values ϕ_0 and $\tilde{\phi}_0$ both produced the same boundary values $\phi(0) = 0$ and $\phi(d) = V_0$, then their difference would solve Laplace's equation with zero boundary values, forcing the difference to be zero everywhere (by uniqueness). Hence $\phi_0 = \tilde{\phi}_0$. So there is **exactly one** interface potential $\phi(x_0)$ compatible with both plates.

Method of Images: Charge near a grounded right-angle conducting corner

Problem setup

Two infinite conducting planes meet at right angles and are grounded:

$$x = 0 : \phi = 0, \quad y = 0 : \phi = 0.$$



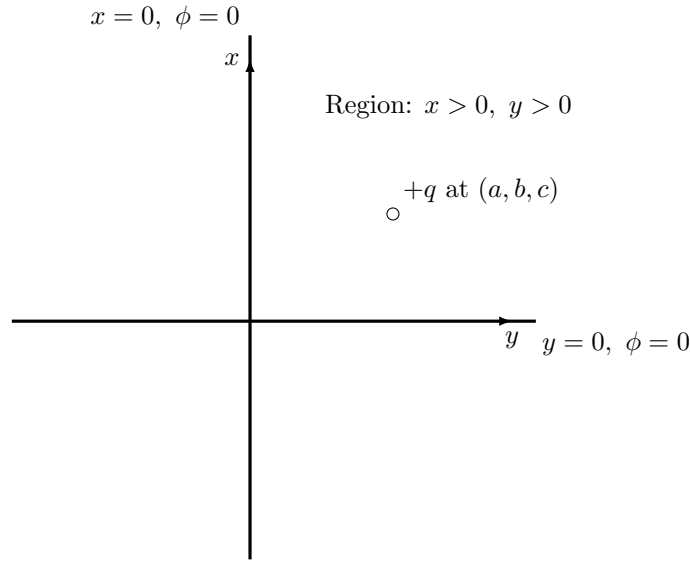


Figure 4: Actual problem: a point charge $+q$ in the region $x > 0, y > 0$ near two mutually perpendicular grounded conducting planes.

A point charge q is located in the region $x > 0, y > 0$ at

$$\mathbf{r}_1 = (a, b, c), \quad a > 0, b > 0.$$

We want the potential $\phi(\mathbf{r})$ and electric field $\mathbf{E}(\mathbf{r})$ in the physical region

$$\Omega : x > 0, y > 0.$$

Step 1: Choose image charges to enforce $\phi = 0$ on both planes

Reflect the real charge across each grounded plane:

$$\mathbf{r}_2 = (-a, b, c), \quad q_2 = -q \quad (\text{mirror across } x = 0),$$

$$\mathbf{r}_3 = (a, -b, c), \quad q_3 = -q \quad (\text{mirror across } y = 0),$$

and reflect across both planes:

$$\mathbf{r}_4 = (-a, -b, c), \quad q_4 = +q.$$

So the original boundary-value problem in Ω is replaced by a **four point-charge** configuration in free space.

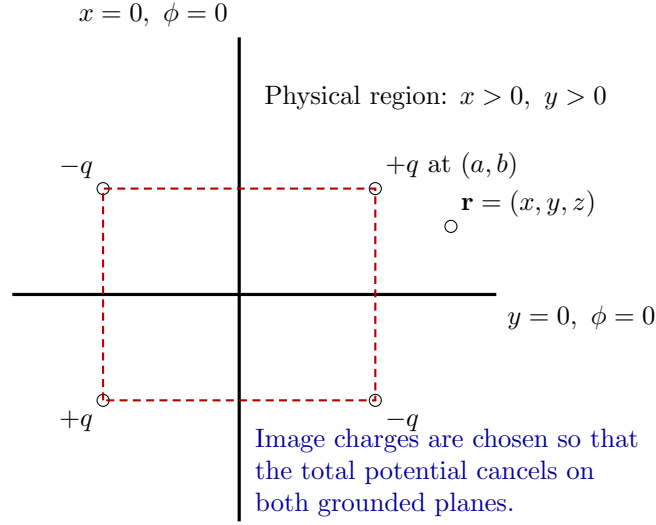


Figure 5: Grounded conducting corner ($x = 0$ and $y = 0$). The image construction replaces the boundary-value problem in Ω by a four-charge free-space problem. (2D top view; the z -coordinate c is unchanged for all images.)

Step 2: Write the potential in Ω by superposition

For any field point $\mathbf{r} = (x, y, z) \in \Omega$, define

$$R_i \equiv |\mathbf{r} - \mathbf{r}_i|, \quad i = 1, 2, 3, 4.$$

Then

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{R_1} - \frac{q}{R_2} - \frac{q}{R_3} + \frac{q}{R_4} \right), \quad x > 0, y > 0.$$

Explicitly,

$$\begin{aligned} R_1 &= \sqrt{(x-a)^2 + (y-b)^2 + (z-c)^2}, \\ R_2 &= \sqrt{(x+a)^2 + (y-b)^2 + (z-c)^2}, \\ R_3 &= \sqrt{(x-a)^2 + (y+b)^2 + (z-c)^2}, \\ R_4 &= \sqrt{(x+a)^2 + (y+b)^2 + (z-c)^2}. \end{aligned}$$

Step 3: Verify the boundary conditions (why this works)

On $x = 0$: When $x = 0$, we have $R_1 = R_2$ and $R_3 = R_4$. Hence

$$\phi(0, y, z) = \frac{1}{4\pi\epsilon_0} \left(q \left(\frac{1}{R_1} - \frac{1}{R_2} \right) - q \left(\frac{1}{R_3} - \frac{1}{R_4} \right) \right) = 0.$$

On $y = 0$: When $y = 0$, we have $R_1 = R_3$ and $R_2 = R_4$. Hence $\phi(x, 0, z) = 0$. Therefore the constructed ϕ satisfies $\phi = 0$ on both grounded planes.

Step 4: Electric field (free-space Coulomb form)

Using superposition (or $\mathbf{E} = -\nabla\phi$), the field in Ω is

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^4 q_i \frac{\mathbf{r} - \mathbf{r}_i}{|\mathbf{r} - \mathbf{r}_i|^3}, \quad (q_1, q_2, q_3, q_4) = (q, -q, -q, +q).$$

Conclusion (method of images)

A difficult boundary-value problem (two grounded conducting planes) is converted into a **free-space superposition problem** by placing **three image charges**. The resulting ϕ satisfies Laplace/Poisson in the physical region and meets the boundary conditions; by the **uniqueness theorem**, it is the unique solution in $x > 0, y > 0$.

A case where the method of images *fails*

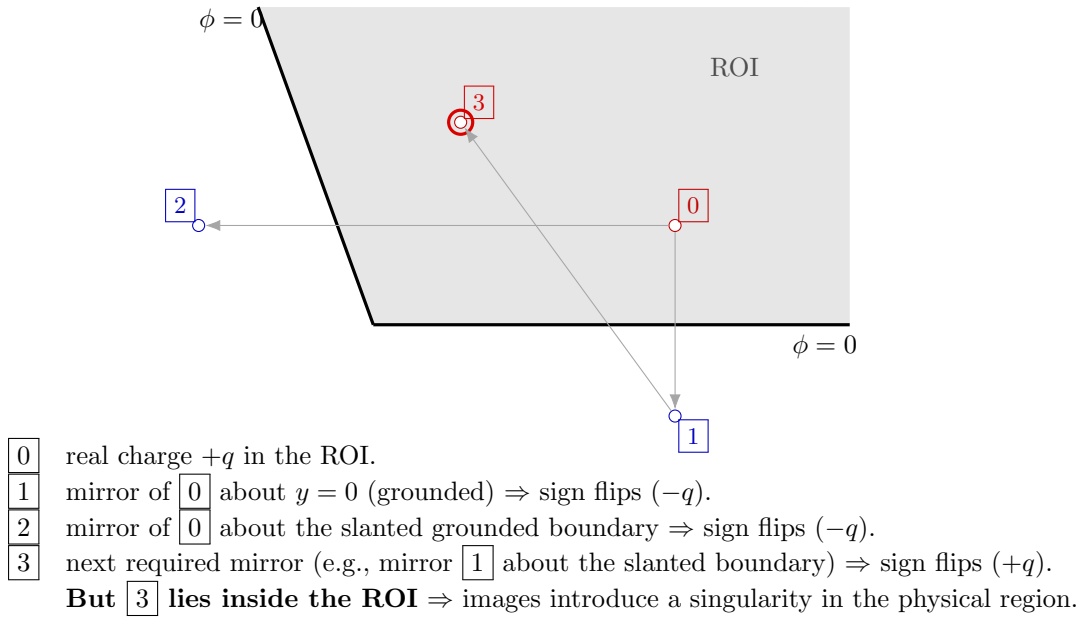


Figure 6: Systematic image-building for two grounded conductors. When the next required image charge appears inside the region of interest (circled), the method of images is invalid.

Why $\boxed{3}$ is the mirror of $\boxed{1}$ about the slanted grounded line (simple geometry)

Think of the slanted grounded conductor as a *mirror line* L (a straight line in the xy -plane).

Step A: How to locate the mirror point (pure geometry). To get the mirror of point $\boxed{1}$ across the slanted line L :

1. Draw a perpendicular from $\boxed{1}$ to the line L . Let the foot of the perpendicular be M .
2. Measure the distance s from $\boxed{1}$ to the line along this perpendicular: $s = |\boxed{1}M|$.
3. Go the same distance s on the *other side* of the line along the same perpendicular. The point you reach is $\boxed{3}$.

So, M is the midpoint of the segment joining $\boxed{1}$ and $\boxed{3}$, and the line L is the perpendicular bisector:

$$|\boxed{1}M| = |\boxed{3}M|, \quad \boxed{1}M \perp L.$$

Step B: Why this guarantees $\phi = 0$ on the slanted grounded line. Pick *any* point P on the slanted boundary line L . Because $\boxed{3}$ is the mirror of $\boxed{1}$, the distances to the line are symmetric, and one gets the key fact:

$$|P - \boxed{1}| = |P - \boxed{3}| \text{ for every } P \in L.$$

Therefore, if we choose opposite charges at these two symmetric points,

$$\boxed{1}: -q, \quad \boxed{3}: +q,$$

their potentials at any boundary point $P \in L$ cancel:

$$\phi(P) \propto \frac{-q}{|P - \boxed{1}|} + \frac{+q}{|P - \boxed{3}|} = (-q + q) \frac{1}{|P - \boxed{1}|} = 0.$$

Hence the pair $\boxed{1}$ and $\boxed{3}$ enforces $\phi = 0$ on the slanted grounded conductor, exactly like a usual mirror-image charge for a single grounded plane.

One-line memory aid

To enforce $\phi = 0$ on a grounded straight conductor: reflect the charge position across the conductor line, and flip the sign.

Why “uniqueness + images” breaks here (what actually fails)

Uniqueness theorem itself does not fail. What fails is the *assumption needed to use it*:

- The true solution ϕ in the region of interest must satisfy

$$\nabla^2 \phi(\mathbf{r}) = -\frac{\rho_{\text{real}}(\mathbf{r})}{\epsilon_0} \quad \text{in the region of interest,}$$

i.e., it has singularities only at the *real* charge locations.

- The method of images is valid only when all *image charges lie outside* the region of interest. Then, inside the region, the constructed $\tilde{\phi}$ satisfies the *same* Poisson/Laplace equation and the same boundary conditions, so uniqueness implies $\tilde{\phi} = \phi$.
- In this example, an image charge appears *inside* the region of interest, so the constructed field satisfies

$$\nabla^2 \tilde{\phi}(\mathbf{r}) = -\frac{\rho_{\text{real}}(\mathbf{r}) + \rho_{\text{image}}(\mathbf{r})}{\epsilon_0} \quad \text{in the region of interest,}$$

which is a **different** source term. Therefore uniqueness cannot be invoked to identify $\tilde{\phi}$ with the true ϕ .

Conclusion: images must never introduce additional singularities in the physical region; otherwise the “image solution” solves the wrong PDE in the region of interest. Such cases are handled using **Green’s functions** (or other BVP methods): A topic for other advanced courses.